Computational Game Theory **Midterm Test 01A** 3 November 2026

Name: _____

Question	1	2	3	4	5	*Total*
Points	6	6	6	6	6	30
Score						

1. **Food trucks**

   Two food trucks simultaneously choose *Station* (12 customers) or *Campus* (8 customers). A truck alone at a spot serves all its customers; two trucks at the same spot split them equally. Payoff = customers served.

   (a) (2 pts) Write the game in normal form.

   (b) (1 pts) Now, whenever the trucks share a spot, Truck 1 gets $\frac{3}{4}$ of the customers and Truck 2 gets $\frac{1}{4}$. Write the new normal form.

   (c) (3 pts) In the new game, find all pure Nash equilibria and all Pareto optimal outcomes. Is every Pareto optimal outcome an equilibrium?

Solution:

(a) Truck 1 chooses rows, Truck 2 columns.

	Campus	Station
Campus	$4, 4$	$8, 12$
Station	$12, 8$	$6, 6$

(b) When the trucks share, Truck 1 serves $\frac{3}{4} \cdot 8 = 6$ or $\frac{3}{4} \cdot 12 = 9$ customers:

	Campus	Station
Campus	$6, 2$	$8, 12$
Station	$12, 8$	$9, 3$

(c) For Truck 1, Station strictly dominates Campus ($12 > 6$, $9 > 8$). Truck 2's best response to Station is Campus ($8 > 3$). The unique pure Nash equilibrium is (Station, Campus) with payoffs $(12, 8)$.

Pareto optimal outcomes: (Campus, Station) = $(8, 12)$ and (Station, Campus) = $(12, 8)$; (Station, Station) = $(9, 3)$ and (Campus, Campus) = $(6, 2)$ are Pareto dominated by $(12, 8)$. So (Campus, Station) is Pareto optimal but not an equilibrium. In the original game both of these outcomes were equilibria; Truck 1's popularity breaks the symmetry in its favour.

2. **Checking claims**

	L	R
U	$4, 1$	$0, 2$
M	$3, 2$	$2, 1$
D	$0, 3$	$4, 2$

In (a), (b) and (c), decide whether the claim is correct and say what you checked.

(a) (1 pts) "(M, L) is a Nash equilibrium, since L is Column's best response to M."

(b) (2 pts) "$(\frac{1}{2}U+\frac{1}{2}D,\ \frac{1}{2}L+\frac{1}{2}R)$ is a Nash equilibrium, since both players are indifferent between the strategies they use." Which condition is missing?

(c) (1 pts) "(D, R) is Pareto optimal, so it is a Nash equilibrium." Check both parts of the claim. How does checking a Nash equilibrium differ from checking Pareto optimality?

(d) (2 pts) Row's equilibrium strategy uses only U and M. Find the Nash equilibrium and verify it.

Solution:

(a) Wrong. Column's response is checked ($2 > 1$), but Row's is not: against L, Row prefers U ($4 > 3$). A Nash equilibrium requires *every* player to best respond. (The game has no pure equilibrium.)

(b) Column: against $(\frac{1}{2}, 0, \frac{1}{2})$, L gives $\frac{1}{2} + \frac{3}{2} = 2$ and R gives $1 + 1 = 2$, indifferent. Row: against $(\frac{1}{2}, \frac{1}{2})$, U and D give 2, but M gives $\frac{3}{2} + 1 = \frac{5}{2} > 2$. Not an equilibrium: Row gains by switching to M. The forgotten condition: strategies *outside* the support must not give more than the strategies in the support.

(c) The first part is true: $(D, R) = (4, 2)$ is Pareto optimal. The only other outcome with Row's payoff 4 is $(U, L) = (4, 1)$, worse for Column, and every outcome better for Column, $(D, L) = (0, 3)$, is worse for Row. The conclusion is false: at (D, R) Column deviates to L ($3 > 2$), so it is not a Nash equilibrium. A Nash equilibrium is checked against *unilateral deviations* of each player, keeping the other's strategy fixed; Pareto optimality is checked against *all other outcomes*, comparing both players' payoffs. Neither property implies the other.

(d) Column indifferent between L and R when Row mixes U and M with $t = P(U)$: $t + 2(1 - t) = 2t + (1 - t)$, so $t = \frac{1}{2}$. Row indifferent between U and M ($q = P(L)$): $4q = 3q + 2(1 - q)$, so $q = \frac{2}{3}$. Check: U and M give $\frac{8}{3}$, D gives $\frac{4}{3} < \frac{8}{3}$. Equilibrium $(\frac{1}{2}U + \frac{1}{2}M,\ \frac{2}{3}L + \frac{1}{3}R)$, payoffs $(\frac{8}{3}, \frac{3}{2})$.

3. **Rock–paper–scissors with a big rock**

Rock beating Scissors scores 2, every other win scores 1. Row's payoffs (zero-sum):

	R	P	S
R	0	-1	2
P	1	0	-1
S	-2	1	0

A student says: "The game is symmetric, so neither player has an advantage. Hence playing R, P, S with probability $\frac{1}{3}$ each is optimal."

(a) (2 pts) Row plays uniformly. Compute Column's expected payoff for each of her pure strategies. Which one exploits Row, and how much does Column gain per round?

(b) (2 pts) Show that $x = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ gives Row the payoff 0 against every pure strategy of Column. Why does this mean that x cannot be exploited?

(c) (1 pts) Why is Paper played most in x, although the big win comes from Rock?

(d) (1 pts) What is wrong with the student's argument? What does symmetry of the game actually imply?

Solution:

(a) Against $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ Row's expected payoffs against the columns R, P, S are $-\frac{1}{3}$, 0, $\frac{1}{3}$, so Column gets $\frac{1}{3}$ with R, 0 with P and $-\frac{1}{3}$ with S. Column exploits the uniform strategy by always playing Rock and wins $\frac{1}{3}$ per round.

(b) Against R: $\frac{1}{2} \cdot 1 + \frac{1}{4} \cdot (-2) = 0$; against P: $\frac{1}{4} \cdot (-1) + \frac{1}{4} \cdot 1 = 0$; against S: $\frac{1}{4} \cdot 2 + \frac{1}{2} \cdot (-1) = 0$. A mixed strategy of Column gives an average of these, so Row gets 0 whatever Column does: there is no reply that makes Row lose. (Column can also guarantee 0 with the same strategy, so the value is 0 and x is optimal.)

(c) The big prize makes Rock attractive to the opponent, and the best answer to Rock is Paper. Scissors is risky (it loses 2 to Rock), and Rock is only needed often enough to keep the opponent from playing Scissors freely. The mix is chosen to make the *opponent* indifferent, not to chase the largest payoff.

(d) Symmetry is a property of the *players*, not of the *actions*: both players face the same game, but R, P, S are not interchangeable. Symmetry implies that the value is 0 (if one player could guarantee a positive payoff, the other could copy his strategy and guarantee the same) and that both players have the same optimal strategies. It does not imply that the optimal strategy is uniform; that holds in ordinary rock–paper–scissors because there the actions are symmetric too.

4. **Correlated equilibrium in Chicken**

Two drivers simultaneously choose to Yield (Y) or Dare (D):

	Y	D
Y	$6,6$	$2,7$
D	$7,2$	$0,0$

Let $\alpha, \beta, \gamma, \delta$ be the probabilities of (Y,Y), (Y,D), (D,Y), (D,D).

(a) (3 pts) Write and simplify the correlated equilibrium conditions. Give one correlated equilibrium.

(b) (3 pts) A device recommends (Y,Y) with probability $\frac{3}{5}$ and (Y,D), (D,Y) with $\frac{1}{5}$ each, and privately tells each driver only his own recommendation. What does each driver get if both obey? Is it a correlated equilibrium? If not, which recommendation is not followed, and what does a deviating driver gain?

Solution:

(a) Driver 1 told Y: $6\alpha+2\beta \geq 7\alpha+0\beta$, i.e. $2\beta \geq \alpha$. Driver 1 told D: $7\gamma+0\delta \geq 6\gamma+2\delta$, i.e. $\gamma \geq 2\delta$. Driver 2 told Y: $6\alpha+2\gamma \geq 7\alpha$, i.e. $2\gamma \geq \alpha$. Driver 2 told D: $7\beta \geq 6\beta+2\delta$, i.e. $\beta \geq 2\delta$. Together with $\alpha, \beta, \gamma, \delta \geq 0$, $\alpha+\beta+\gamma+\delta = 1$.

Examples of correlated equilibria (any one suffices): the pure Nash equilibrium (Y, D), i.e. $\beta = 1$; the "traffic light" $\beta = \gamma = \frac{1}{2}$; or $\alpha = \beta = \gamma = \frac{1}{3}$, $\delta = 0$. Each satisfies all four inequalities.

(b) If both obey, each driver gets $\frac{3}{5}\cdot 6 + \frac{1}{5}\cdot 2 + \frac{1}{5}\cdot 7 = \frac{27}{5}$. Not a correlated equilibrium: the condition $2\beta \geq \alpha$ fails ($\frac{2}{5} < \frac{3}{5}$). A driver told Y knows the other was told Y with probability $\frac{3/5}{4/5} = \frac{3}{4}$ and D with probability $\frac{1}{4}$: obeying gives $\frac{3}{4}\cdot 6 + \frac{1}{4}\cdot 2 = 5$, daring gives $\frac{3}{4}\cdot 7 + \frac{1}{4}\cdot 0 = \frac{21}{4}$. So he ignores the recommendation Y and dares, gaining $\frac{1}{4}$ whenever he is told Y. (A driver told D knows the other yields; $7 > 6$, so he obeys.)

5. **Guard and thief**

A guard patrols warehouse A (loot 4) or B (loot 2). A thief robs A, robs B, or stays home (H). Robbing the patrolled warehouse: thief caught, thief -3, guard $+1$. Robbing the other one: the thief takes the loot, the guard loses its value. Staying home: both 0.

The guard commits to $p = P(\text{patrol } B)$; the thief sees p (not the patrol) and best responds.

(a) (2 pts) Write the game in normal form.

(b) (2 pts) For each thief action, write the linear program for the guard's best p, given that this action is the thief's best response.

(c) (2 pts) Solve the programs and find the strong Stackelberg equilibrium. Does a weak one exist?

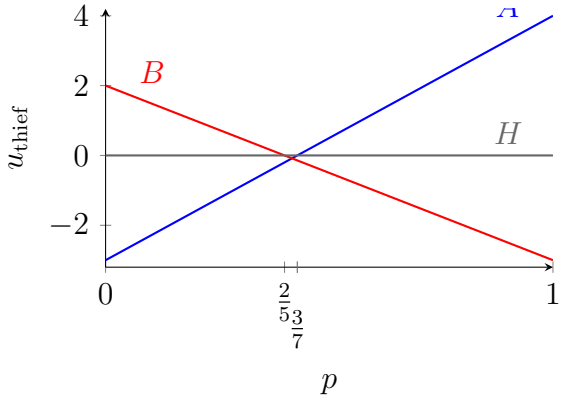
Solution:

(a) Guard (rows) and thief (columns):

	rob A	rob B	H
patrol A	$1, -3$	$-2, 2$	$0, 0$
patrol B	$-4, 4$	$1, -3$	$0, 0$

(b),(c) With $p = P(\text{patrol } B)$ the expected payoffs are

	A	B	H
u_{thief}	$7p - 3$	$2 - 5p$	0
u_{guard}	$1 - 5p$	$3p - 2$	0



$$LP_A: \max\ 1 - 5p \ \text{ s.t. } 7p - 3 \geq 2 - 5p,\ 7p - 3 \geq 0 \quad \Rightarrow p \geq \tfrac{3}{7},\ \text{value } -\tfrac{8}{7},$$
$$LP_B: \max\ 3p - 2 \ \text{ s.t. } 2 - 5p \geq 7p - 3,\ 2 - 5p \geq 0 \quad \Rightarrow p \leq \tfrac{2}{5},\ \text{value } -\tfrac{4}{5},$$
$$LP_H: \max\ 0 \ \text{ s.t. } 0 \geq 7p - 3,\ 0 \geq 2 - 5p \quad \Rightarrow \tfrac{2}{5} \leq p \leq \tfrac{3}{7},\ \text{value } 0.$$

The best program is LP_H: in a strong Stackelberg equilibrium the guard patrols B with any probability $p \in [\frac{2}{5}, \frac{3}{7}]$, the thief stays home, and both get 0. Deterrence: the guard never needs to catch anyone.

A weak Stackelberg equilibrium exists here (unlike Exercises 5.27 and 5.28): for any p strictly between $\frac{2}{5}$ and $\frac{3}{7}$ the thief strictly prefers H, so the way ties are broken does not matter.